With two rounds of integration by parts on the term involving $b(x)$, we find, using the fact that the boundary terms vanish,

$$\int_{-\infty}^{+\infty} b^2(x)\phi''(x)\rho(x,t)\mathrm{d}x = \int_{-\infty}^{+\infty} \phi(x)(b^2(x)\rho)_{xx}(x,t)\mathrm{d}x.$$

Finally, putting these results together, we have

$$\int_{-\infty}^{+\infty} \phi(x)\left[\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x}(a(x)\rho(x,t)) - \frac{1}{2}\frac{\partial^2}{\partial x^2}(b^2(x)\rho)\right]\mathrm{d}x = 0.$$

If this is to hold for all test functions ϕ, we must have

$$\frac{\partial\rho}{\partial t} = -\frac{\partial}{\partial x}(a(x)\rho) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(b^2(x)\rho).$$

This partial differential equation is often called the Fokker-Planck equation. The great benefit of the Fokker-Planck equation is that it enables the macroscopic perspective on the solutions of an SDE.

The Fokker-Planck equation may be written in the compact form

$$\frac{\partial\rho}{\partial t} = \mathcal{L}^\dagger\rho,$$

where the operator appearing on the right hand side is the *adjoint* $\mathcal{L}^\dagger$ of the generator of the stochastic process, satisfying the relation

$$\langle\mathcal{L}\phi,\rho\rangle = \langle\phi,\mathcal{L}^\dagger\rho\rangle.$$

In the case at hand the adjoint is

$$\mathcal{L}^\dagger\rho = -(a(x)\rho)_x + \frac{1}{2}(b^2(x)\rho)_{xx},$$

which we shall refer to as the Kolmogorov operator.

In case $b \equiv 0$, the SDE becomes an ordinary differential equation. Then the generator $\mathcal{L}$ becomes simply the Lie derivative, and its adjoint is just the Liouvillian. Thus the framework developed here for SDEs is a proper generalization of what we have earlier seen in the deterministic case.